BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 139 · Operating Systems

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Give a physical security measure and the threat it limits.

Q2. What is penetration testing?
A. An attack without permission
B. Deleting all user accounts
C. Security testing with permission
D. Compressing every stored file

Q3. “POP's main job is sending a new message from a client to a mail server.” Correct this. Explain why.

Q4. A pupil wants matching inboxes and read status on a phone and laptop. “IMAP only sends new outgoing mail.” Correct this. Why use IMAP here?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

User interface — CLI uses typed commands; GUI uses graphics and controls.

Memory management — OS allocates and manages memory for active processes.

Multitasking — OS schedules CPU time among tasks/processes.

File and peripheral management — OS organises files and controls device access.

New learning

User interface: CLI uses typed commands; GUI uses graphics and controls.

Memory management: OS allocates and manages memory for active processes.

Multitasking: OS schedules CPU time among tasks/processes.

File and peripheral management: OS organises files and controls device access.

Worked example

A laptop plays music while a document is edited. Explain three operating-system responsibilities.

1. Allocate CPU processing time to both programs.
2. Allocate memory for active processes.
3. Manage access to audio hardware and saved document files.
4. Describe how scheduling gives an impression of simultaneous progress.

We Do · Practise with guidance

Guided 1

Compare use of a CLI and GUI to manage school computers.

Worked method: Both allow users to interact with OS services.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete an assignment showing how much RAM remains after a program loads.

```
freeRAM = 900
usedRAM = 300
remaining = ...
print(remaining)
```

write

Question 1

Name two operating-system functions. [2 marks]

Question 2

Explain process scheduling in multitasking. [3 marks]

Question 3

Give one benefit of GUI over CLI. [2 marks]

Exit Ticket · Answer before checking

An OS manages processes, files, memory and devices.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Lock the server room to limit access and reduce theft or tampering.

Q2. C — Security testing with permission

Q3. POP downloads email from a server to a device. IMAP keeps email views in sync across devices.

Q4. IMAP keeps mail on the server and syncs messages and read status across devices. SMTP sends outgoing mail.

I Do / We Do pages · Guided practice

Q1. Both allow users to interact with OS services.

You Do pages · Code First

```
freeRAM = 900
usedRAM = 300
remaining = freeRAM - usedRAM
print(remaining)
```

Expected result: 600

You Do pages · Independent questions

Q1. Memory management, process scheduling, file management, user interface or peripherals (any two).

Q2. The OS allocates CPU time to processes so multiple programs can make progress.

Q3. Many users find labelled visual controls easier to learn/use.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: